

(3.2)

a) Označme: $\vec{v}_1 := 1$

$$\vec{v}_2 := x$$

$$\vec{v}_3 := x^2$$

Provedeme Gram-Schmidtovu ortogonalizaci na bázi $(\vec{v}_1, \vec{v}_2, \vec{v}_3)$, čímž získáme posloupnost $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$, která bude ON bází podprostoru $\text{Span}\{1, x, x^2\}$

$$\vec{u}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$$

$$\|\vec{v}_1\| = \sqrt{\langle 1, 1 \rangle} = \sqrt{\int_1^5 1 \, dx} = \sqrt{5-1} = 2$$

$$\vec{u}_1 = \frac{1}{2}$$

$$\vec{u}_2' = \vec{v}_2 - \langle \vec{u}_1, \vec{v}_2 \rangle \vec{u}_1 = x - \left\langle \frac{1}{2}, x \right\rangle \frac{1}{2}$$

$$\left\langle \frac{1}{2}, x \right\rangle = \int_1^5 \frac{1}{2} x \, dx = \left[\frac{x^2}{4} \right]_1^5 = \frac{25}{4} - \frac{1}{4} = 6$$

$$\vec{u}_2' = x - 3$$

$$\|\vec{u}_2'\| = \sqrt{\int_1^5 (x^2 - 6x + 9) \, dx} = \sqrt{\left[\frac{x^3}{3} - 3x^2 + 9x \right]_1^5} = \sqrt{\frac{125}{3} - 75 + 45 - \frac{1}{3} + 3 - 9} = \sqrt{36 + 41 + \frac{1}{3}} = \sqrt{\frac{116}{3}} = \frac{4}{\sqrt{3}}$$

$$\vec{u}_2 = \frac{\sqrt{3}}{4}(x-3)$$

$$\vec{u}_3' = \vec{v}_3 - (\langle \vec{u}_1, \vec{v}_3 \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v}_3 \rangle \vec{u}_2) = x^2 - \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{2} - \left\langle \frac{\sqrt{3}}{4}(x-3), x^2 \right\rangle \left(\frac{\sqrt{3}}{4}(x-3) \right)$$

$$\left\langle \frac{1}{2}, x^2 \right\rangle = \int_1^5 \frac{x^2}{2} \, dx = \left[\frac{x^3}{6} \right]_1^5 = \frac{5^3}{6} - \frac{1^3}{6} = \frac{124}{6} = \frac{62}{3}$$

$$\begin{aligned} \left\langle \frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}, x^2 \right\rangle &= \int_1^5 \left(\frac{\sqrt{3}}{4}x^3 - \frac{3\sqrt{3}}{4}x^2 \right) \, dx = \left[\frac{\sqrt{3}}{16}x^4 - \frac{\sqrt{3}}{4}x^3 \right]_1^5 = \frac{5^4\sqrt{3}}{16} - \frac{5^3\sqrt{3}}{4} - \left(\frac{\sqrt{3}}{16} - \frac{\sqrt{3}}{4} \right) \\ &= \frac{5^4\sqrt{3}(5-4)}{16} - \frac{\sqrt{3} - 4\sqrt{3}}{16} = \frac{128}{16}\sqrt{3} - 8\sqrt{3} \end{aligned}$$

$$\vec{u}_3' = x^2 - \frac{62}{3} \cdot \frac{1}{2} - 8\sqrt{3} \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) = x^2 - \frac{31}{3} - 6x + 18 = x^2 - 6x + \frac{23}{3}$$

$$\|\vec{u}_3'\| = \sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3} \right)^2 \, dx} = \sqrt{\frac{256}{45}} = \frac{16}{3\sqrt{5}}$$

$$\vec{u}_3 = \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3} \right) = \frac{\sqrt{5}}{16} (3x^2 - 18x + 23)$$

Hledanou ON bází je tedy $\left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{\sqrt{5}}{16}(3x^2 - 18x + 23) \right)$

- b) Dle tvrzení 8.54 lze OG projekce $\vec{w}$ vektoru $\vec{v}$ na podprostor generovaný ON bází $(\vec{u}_1, \dots, \vec{u}_k)$ spočítat jako
- $$\vec{w} = \langle \vec{u}_1, \vec{v} \rangle \vec{u}_1 + \dots + \langle \vec{u}_k, \vec{v} \rangle \vec{u}_k$$

Máme ON bázi $B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{\sqrt{5}}{16}(3x^2-18x+23) \right)$

Označme $\vec{v} := \sin x$

Potom platí:

$$\vec{w} = \left\langle \frac{1}{2}, \sin x \right\rangle \frac{1}{2} + \left\langle \frac{\sqrt{3}}{4}(x-3), \sin x \right\rangle \frac{\sqrt{3}}{4}(x-3) + \left\langle \frac{\sqrt{5}}{16}(3x^2-18x+23), \sin x \right\rangle \frac{\sqrt{5}}{16}(3x^2-18x+23)$$

$$\left\langle \frac{1}{2}, \sin x \right\rangle = \int_1^5 \frac{1}{2} \sin x \, dx = \frac{-\cos 5 + \cos 1}{2} \approx 0,128$$

$$\left\langle \frac{\sqrt{3}}{4}(x-3), \sin x \right\rangle = \int_1^5 \frac{\sqrt{3}}{4}(x-3) \sin x \, dx \approx -1,493$$

$$\left\langle \frac{\sqrt{5}}{16}(3x^2-18x+23), \sin x \right\rangle = \int_1^5 \frac{\sqrt{5}}{16}(3x^2-18x+23) \cdot \sin x \, dx \approx -0,125$$

$$\vec{w} \approx 0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{\sqrt{3}}{4}x - 1,493 \cdot \frac{\sqrt{3}}{4} \cdot 3 - 0,125 \cdot \frac{\sqrt{5}}{16} \cdot 3x^2 + 0,125 \cdot \frac{\sqrt{5}}{16} \cdot 18x - 0,125 \cdot \frac{\sqrt{5}}{16} \cdot 23$$

$$\vec{w} = x^2 \left(-0,125 \cdot \frac{\sqrt{5}}{16} \cdot 3 \right) + x \left(-1,493 \cdot \frac{\sqrt{3}}{4} + 0,125 \cdot \frac{\sqrt{5}}{16} \cdot 18 \right) + \left(0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{3\sqrt{3}}{4} - 0,125 \cdot \frac{\sqrt{5}}{16} \cdot 23 \right)$$

$$\approx -0,052 \quad \approx -0,332 \quad \approx 1,602$$

Hledaná OG projekce je tedy (přibližně):

$$\vec{w} \approx -0,052x^2 - 0,332x + 1,602$$

- c) Norma vektoru $\vec{u}$ se spočítá jako $\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$

Tedy:

$$\|\sin x + 0,052x^2 + 0,332x - 1,602\| = \sqrt{\int_1^5 (\sin x + 0,052x^2 + 0,332x - 1,602)^2 dx}$$

$$\approx \sqrt{0,101744}$$

$$\approx 0,319$$

Norma rozdílu je tedy přibližně 0,319.

(Je to malé číslo, což se dalo očekávat, protože OG projekce je nejlepší aproximací vektoru v tom smyslu, že minimalizuje právě normu tohoto rozdílu.)